

Smaller and Shallower XOR Circuits for AES MixColumns: 97 Gates at Depth 3; 92 at Depth 4; 89 at Depth 5; 88 at Depth 6

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Abstract

We give seven verified implementations of AES MixColumns as circuits of 2-input XOR gates over $\text{GF}(2)$. Four improve the published depth–count Pareto frontier at their depth, on lineages holding no imported circuit: **97** gates at depth **3**, the minimum depth, against 99 [2]; **92** at depth **4** against 97 [3]; **89** at depth **5** against 94 [3]; and **88** at depth **6** against 92 [4, 5], one level shallower than Jean’s published 88 [9] at the same count. **88 is not a new count**: it is Jean’s, and **Jean has priority**. A fifth circuit, 88 at depth 7, *ties* that point with an independent circuit sharing 61 of 88 internal masks. A sixth, 88 at depth 5, is two levels shallower again and six gates below the published depth-5 point, but its seed chain passes through Jean’s circuit and is reported as derived work; so is the seventh, 88 at depth 8, which is dominated and claims nothing. Combined frontier **97 @ 3, 92 @ 4, 88 @ 5**; with no imported material, **97 @ 3, 92 @ 4, 89 @ 5, 88 @ 6**. All were found by stochastic local search over a value-set formulation of the shortest-linear-program problem. No 87 was found: 47 canonical 88-gate circuits, and the new 88 at depth 6, have exhaustively empty remove- ≤ 3 neighbourhoods. Nothing here is claimed optimal.

1 Model and specification

A circuit is an ordered list of 2-input XOR gates: $s_0, \dots, s_{31}$ are the input bits, and gate k produces $s_{32+k} = s_a \oplus s_b$ with $a, b < 32 + k$. Cost is the gate count; depth counts gates on the longest input-to-output path, inputs at depth 0. Bit index i denotes bit $(i \bmod 8)$ of state byte $\lfloor i/8 \rfloor$, least-significant-bit first, and MixColumns maps an input column $a_0, \dots, a_3$ to

$$\text{out}[c] = 2 \cdot a_c \oplus 3 \cdot a_{(c+1) \bmod 4} \oplus a_{(c+2) \bmod 4} \oplus a_{(c+3) \bmod 4},$$

in $\text{GF}(2^8)$ modulo $x^8 + x^4 + x^3 + x + 1$ (0x11b), per NIST FIPS 197-upd1, Eq. 5.6 [1]. The map is $\text{GF}(2)$ -linear, so agreement on the 32 unit-input vectors is a complete correctness check; counts and depths are relabeling-invariant, so no comparison below depends on convention.

The s-XOR (in-place) model is comparable: SSA-unrolling a k -instruction s-XOR program yields exactly k free-standing gates. Multi-input XOR gates, gate-equivalent area and quantum CNOT counts are different cost functions: not comparable, not claimed against.

2 Results

Depth 3 is the known minimum depth — an output depending on w inputs needs depth $\geq \lceil \log_2 w \rceil$, and MixColumns has outputs of weight 7 [2] — so the depth-3 circuit’s contribution

Depth	Published best	Model	This work	Relation
3	99 [2]	XOR2	97	−2 (depth 3 is the minimum depth)
4	97 [3]	XOR2	92	−5
5	94 [3]	XOR2	89	−5
5	94 [3]	XOR2	(88) [†]	−6, but derived from [9]
6	92 [4, 5]	XOR2 / s-XOR	88	−4; dominates the 88 of [9]
7 [‡]	88 [9]	XOR2	88	ties it; independent, 61/88 masks shared
8	—	XOR2	(88) [†]	derived from [9]; dominated
9 [‡]	89 [8]	XOR2	—	off the frontier

Table 1: The published depth–count landscape in comparable models, and this work. 88 is *not* a new count: it is Jean’s [9], who has priority; the depth-6 row improves the depth at that count, not the count. [†]Derived: the seed chain passes through [9], which is credited; not an independent construction. [‡]*Neither* [9] *nor* [8] states a depth: the 7 and the 9 are both our own measurements of our own transcriptions, on the same footing, and both are *forced* — the ASAP least-fixpoint schedule over each published mask set, the shallowest either admits, still gives 7 and 9. [8] reports its count as “g-XOR”, which is the same model as the XOR2 of every other row (Section 1). Jean’s 88 [9] supersedes the earlier 91 of [6, 7] at depth 7; the depth-3 lineage is 105 → 103 → 102 → 99 ([2], Table 3). Listings: Appendices A–G.

is its count. The 89 at depth 5 is shallower than any published circuit of fewer than 94 gates whose depth is stated, and stays on the published frontier: the published sub-89 point, Jean’s 88 [9], measures depth 7 (Table 1, note [‡]). Within this note it is now undercut at its own depth, but only by derived work (below), so it remains the depth-5 point of what we reached with nothing imported.

The 88 at depth 7 *ties* Jean’s point rather than improving it: Jean’s note was posted 2026-07-23 and has priority. Ours was found 2026-07-26 on a fully logged own lineage — from-scratch 97 at depth 3 (235s), a 21 h ladder to 89 at depth 6 (11.0 h), one reroute to 89 at depth 5 (592s), ρ^2 -symmetrization to a 94 at depth 5, then 32.9 min of plateau walking — with no imported circuit in the chain, and is a different circuit: 61 of 88 masks shared, Jaccard 0.530. For scale, Jean’s 88 and the 89 of [8] — two independently published circuits — share 63; overlap of that size is what independence looks like for this map. The 88 at depth 8 (Appendix G) is derived work and dominated: its seed chain passes through Jean’s, ρ^2 -symmetrized and peeled to 95, orbit-walked to 92, then unioned with a 91 of our own lineage before the descent.

Two further 88s differ in exactly the way that matters. The **88 at depth 6** (Appendix E), found 2026-07-28, is from scratch: worker `c_naive`, restart 18, rooted in randomized balanced XOR trees over the 32 raw inputs, then 89 → 88 at depth 7 (iteration 37 270) → 88 at depth 6 (iteration 37 501), half a second later on the same depth tie-break. The reason nothing imported can have reached it is the *root*, not a knob setting: that worker’s root spec was `constructor:naive`, and the constructor branch of the archived worker’s root generator calls only `constructors.build(name, seed)` — it contains no file-reading path at all, while every branch that can open a circuit off disk is a different branch, unreachable under this spec. All 38 restarts logged in the session that produced this circuit opened on such a root, and the one that produced it rebuilds to exactly the logged 139-gate depth-3 starting circuit from its seed. Knobs close the remaining routes — cross-pollination is never enabled anywhere in the archived worker *and* the log carries none of the lines the engine emits when a merge brings material in; repulsion was off; each restart carries its own context — but the knobs alone would not settle it, as the derived 88 at depth 5 below shows: its worker had identical knobs, and imported material still reached it, through its root. It is the first 88 we have found *from scratch* — the depth-7 one above is on our own lineage and independent of Jean’s, but was

warm-started from our own earlier records — and it dominates Jean’s: equal count, one level shallower, while undercutting the published depth-6 92 [4, 5] by four gates. The depth 7 that comparison rests on is our own measurement of our own transcription (Table 1, note ‡), but it is *forced*: the ASAP least-fixpoint schedule over Jean’s own mask set — the shallowest schedule any circuit on that mask set admits — still places three of its output bits at depth 7, so that circuit cannot be rescheduled shallower. Its largest mask overlap with any other circuit in the artifact repository is 47 (Jaccard 0.362, against the superseded 89 at depth 10); against any other 88 it is 43 (Jaccard 0.323).

The **88 at depth 5** (Appendix D), found 2026-07-29, is **derived work**, and that is the first thing to say about it: its seed chain passes through Jean’s 88 — symmetrized and peeled to 95, orbit-walked to 92, unioned with a 91 of our own lineage, descended to the exactly ρ^2 -symmetric 90 at depth 9 of Section 4, reseeded mask-identically into the orbit hunt, then polished to 88 at depth 6 and 88 at depth 5, with mask identity checked at every link. It is two levels below the depth-6 88 and six gates below the published depth-5 94 [3]. Neither is it a new construction: 75 of its 88 masks are the depth-5 89’s (Jaccard 0.735, the largest overlap between any two circuits here), so it is that 89’s own basin reached at 88 gates, not a new family.

Two frontiers therefore, and we report both: **97 @ 3, 92 @ 4, 88 @ 5** across all seven circuits, and **97 @ 3, 92 @ 4, 89 @ 5, 88 @ 6** once imported material is excluded. **87 was not found**.

3 Method

A circuit is represented as a *value set*: masks over the 32 inputs, valid iff every mask is the XOR of two already-available masks and all 32 output masks are present; gate count is the set size. Deletion becomes a set operation, circuits compare by mask overlap regardless of gate order, and a neighbourhood — remove k masks, add at most $k - 1$ — becomes finite and enumerable (Section 4).

Good circuits at this size *look* locally efficient — every gate reads as an optimal step — and that is what makes them hard to improve: they sit at local optima, so smaller circuits are separated from them by locally *suboptimal* steps. (This was true of the naive construction we started from and of every circuit we produced up to the 89; we make no claim about how anyone else’s circuit was built.) **Remove-1** drops a non-output mask if the set stays realizable. The **neutral swap** removes one mask and adds one repair mask at unchanged size: with $A = \text{closure}(S')$ and $\text{stuck} = (S' \cup T) \setminus A$, every single-mask repair lies in $C = \{u \oplus a : u \in \text{stuck}, a \in A\} \cap P_2(A)$ (P_2 = pair sums), so enumerating C *decides* the move — measured $|C| \approx 7$, cheaper than the earlier engine’s 24–40 random candidates, which missed 81–82% of existing repairs. **Remove-2-add-1** remains a diversifier, though at the 88/89 frontier it is provably empty (Section 4). **Destroy-and-rebuild** removes a group and re-synthesizes from a pool, accepting rebuilds no larger, or larger on an annealing schedule with reheat; destroying a *connected cone* of the DAG leaves a hole worth re-planning, where k unrelated masks are usually just re-derived, and near-miss rebuilds are peeled before judgement. A fixpoint over minimum build depths rejects cap violations and emits gate lists in depth order.

Workers run one engine at one depth cap in fixed-length chunks, alternating walk and rebuild, and a coordinator re-offers each the best circuit feasible at its cap (the reseeding ladder). A worker accepts fewer gates *or equal gates at strictly lower depth* — a Pareto tie-break, and not a cosmetic one: 89 at depth 6 became 89 at depth 5 through it; both 88s of version 2, first seen at depths 11 and 10, reached depths 7 and 8 seconds later; and the 88 at depth 6 followed its own depth-7 predecessor half a second later. Every candidate is verified against MixColumns rebuilt from $\text{GF}(2^8)$ before acceptance. The depth-3 circuit comes from

a separate constructor: each output is modelled as $A \oplus B$, A and B partitioning its bits into shared, recount-costed depth- ≤ 2 parts, annealed by iterated local search, so depth ≤ 3 holds by construction.

The v2 engine. Version 1’s circuits came from an engine whose kernel was then profiled and rewritten [10]: level-BFS depth relaxation ($17.5\times$ on destroy-and-rebuild, bit-identical trajectories); a worklist closure with incremental removal queries, and removal of a bug that re-verified unchanged circuits every plateau iteration — 78% of walk runtime ($21.9\times$ on the walk); victim repooling ($\sim 34\times$ accepted moves/s); cone destroys ($\sim 12\times$); exact repair ($\sim 7\times$ plateau mobility). The walk now runs at 480–640 iterations/s against 70–110. What found both 88s was not a cleverer move but a cheap plateau walked at volume: every distinct equal-best mask set is harvested.

4 Exact local certificates, and structure

To say what is *not* there we decide the neighbourhood question: given a verified 88-gate circuit and a window of k removed masks, can at most $k - 1$ new masks — each buildable when added, with cascade unlocks between — restore all 32 targets? A “yes” is an 87-gate circuit; a completed “no” is a proof for that window. Completeness follows from a first-unlock case analysis: the first unlock must have the newest added mask as a parent, which pins each added mask to a finite, enumerable shape. Runs above the proved budget are recorded as undecided, never as irreducible. The procedure agrees with an independent brute force 25/25 and 12/12 at $k \leq 3$, and on 1257 instances including 122 genuine negatives at $k = 4$.

- **47 canonical 88-gate circuits** — Jean’s 88, its 12 plateau siblings, and a third family’s anchor with 33 representatives — **have exhaustively empty $k \leq 3$ shells**: all 1540 windows at $k = 2$ and all 27720 at $k = 3$ each, none reducible. Hence *any 87-gate circuit differs from each of the 47 by at least 4 masks*. Stated plainly: all 47 lie in Jean’s-lineage families — families 1–2 are Jean’s circuit and its siblings, and the third family’s anchor is itself seeded through Jean’s circuit — so this is a rigidity statement about *that* neighbourhood, and our 88 at depth 7, independent of that lineage, is exactly the least-certified: it is *not* among the 47, its exhaustive $k \leq 3$ sweep was never run, and it has only 9 exact $k = 4$ windows and 8 SAT cone windows. Version 2 of this note recorded that the best-certified shells and the independent one were therefore disjoint; that no longer holds. **The 88 at depth 6 — from scratch, outside Jean’s lineage — has an exhaustively empty $k \leq 3$ shell of its own** (all 1540 windows at $k = 2$, all 27720 at $k = 3$), so any 87 differs from it too by at least 4 masks. The depth-5 88’s $k = 2$ shell is empty as well (1540 windows, so any 87 differs from it by at least 3 masks), but its $k = 3$ shell was *never swept*: with the depth-7 88 it is one of the two least-certified circuits here.
- **Population.** Harvesting mapped ≈ 139878 distinct 88-gate mask sets, of which 105801 are proven irreducible at $k = 2$. In total ≈ 165 million exact window decisions, zero timeouts, zero reducible windows; all five 89-gate seeds likewise have empty $k = 2$ shells.
- **A retracted observation, and the certificate that survives it.** Version 2 of this note observed that no 88-gate mask set of minimum build depth ≤ 6 had been seen in the harvesting. The two circuits added here, at depths 6 and 5, refute that observation — which was reported as an observation and not as a certified claim. What was certified is narrower and stands: near the *old* 88-plateau no 88 at depth 6 exists, all 4861 two-critical cores being UNSAT at cap 6. Both new circuits live in a different basin, and the

basins differ in exactly the relevant place: the old plateau’s depth obstruction is output rows 3 and 27, the depth-6 88’s is rows 1, 11, 17 and 25 — four weight-7 targets, three of them column-rotations of one another. MixColumns has at least two structurally distinct ways to hit a depth wall at 88 gates, and the second one gives way at depth 6.

- **And no further: the depth-6 basin does not reach depth 5.** The natural next question is whether the depth-6 88’s own family contains a member realizable at depth 5, which would give a Jean-independent circuit at the combined frontier’s depth-5 point. It does not. All 8 993 members of that family were screened by their ASAP (least-fixpoint) depth — the shallowest depth each mask set admits at all — and the histogram is exactly 4 420 at depth 6 and 4 573 at depth 7, with *nothing below 6*: depth 6 is minimal for every member, not merely attained. A depth-capped-5 SAT sweep of 27 windows around the circuit returned 22 UNSAT and **0 SAT**, the 5 undecided ones all at the widest radius. So the derived depth-5 88 remains the only 88 at depth 5 here; a Jean-independent one, if it exists, is not in this basin.
- **Windowed SAT is evidence, not proof.** A sound windowed CNF encoding, solved by Kissat under two soundness-proved symmetry-breaking layers, returned UNSAT up to $k = 16$ on Jean’s 88 and $k = 15$ on the third-family anchor (whose whole $k \leq 12$ regime is UNSAT, 28/28), with **0 SAT anywhere** — but UNSAT there is relative to the encoding’s fixed slot order, and undecided windows are undecided.

None of this bounds 87 away globally; it is consistent with 88 being locally rigid, and it redirected the search: at 88 gates every move class small enough to enumerate is empty, so what remains is walking the equal-size plateau widely.

Symmetry. Byte rotation ρ (order 4) commutes with MixColumns, so the 32 targets form 8 ρ -orbits of size 4 and equivariant circuits can be searched in orbit space — but full ρ -symmetry costs about +19 gates. The half-rotation ρ^2 is where the structure is: in the elite basin 79–81 % of a circuit’s masks have their ρ^2 -image also in the mask set, and the 12 ρ^2 -fixed masks the search settles on are exactly the classic $x_0 \oplus x_2 / x_1 \oplus x_3$ pairs, rediscovered rather than imposed. Jean’s 88, at 75 % (66 of 88), symmetrizes into a second, distinct basin. A ρ^2 -equivariant orbit engine produced two *exactly* ρ^2 -symmetric 90-gate circuits (depths 9 and 7), improving the best exactly symmetric circuit we know of from 94 to 90; both are certified locally optimal in orbit space, and both are derived work, each seed union containing a circuit symmetrized from Jean’s 88; the depth-9 one is the ancestor of the 88 at depth 5 (Section 2). Symmetry is also a seeding device: the exactly ρ^2 -symmetric 94 at depth 5 — 82 of its 94 masks shared with the 89 at depth 5 — costs +5 over that 89 and is the seed from which the 88 at depth 7 was found.

5 Verification, artifacts, and reproduction

Every circuit here is checkable from the note alone: the appendices list every gate, Section 1 defines the target map, and because MixColumns is linear, replaying a listing on the 32 unit-input vectors against Eq. 5.6 [1] is a complete correctness check. Each appendix also states its circuit’s canonical SHA-256 over the UTF-8 bytes of `{"inputCount":32,"gates":[...]}`.

The artifacts automate that. Circuits and verifiers are at <https://github.com/Joe-b-20/aes-mixcolumns-xor-circuits>, archived at concept DOI 10.5281/zenodo.21299092; `python3 verify_all.py` runs `verify.py`, which rebuilds the target map from the $\text{GF}(2^8)$ definition so the verifier *is* the specification, a clean-room verifier with adversarial rejection tests, and Verilog testbenches, all under continuous integration. The method, pipeline, untouched

run archives with each run’s exact code, certificate summaries and the credited transcriptions of the imported circuits [9, 8] are published separately [10], with a reproduction command for every result that has one — the 88 at depth 6 has none, and the derived 88s at depths 5 and 8 are deliberately offered as archives rather than recipes. Measured there on 2026-07-27 with the shipped v2 engine, and reported as measurements rather than promises: the from-scratch 97 at depth 3 in 81s on one core; the 89 at depth 5 in 19s and 22s from shipped seeds, against 592s for the archived 2026-07-14 run on the v1 engine; the 88 at depth 7 in 19.4 and 31.0 minutes in two single-worker re-runs from the ρ^2 -symmetric 94, against 32.9 min in the archived ten-worker run; the 92 at depth 4 a from-scratch cascade of hours. The re-runs are re-runs, not independent confirmations.

Implementation and execution used LLM coding agents as programming tools, directed by the author, who designed the moves and acceptance rules. Concretely: the 97 at depth 3, the 92 at depth 4 and the 89 at depth 5 are reproduced by the published, dependency-free code with no AI system in the loop, as is a single-worker re-run of the 88 at depth 7. *All four* 88-gate circuits came out of author-directed LLM-agent campaigns, in two chapters — the v2 engine, the certificates of Section 4 and the 88s at depths 7 and 8 from a 24-agent campaign over 2026-07-26/27, and the 88 at depth 6 and the 88 at depth 5 from a later multi-day fleet of sixteen processes running that engine, built and operated by the same agents, on 2026-07-28 and 2026-07-29 respectively. The 88 at depth 6 is the one frontier record of this note with *no* single-command reproduction: it came out of restart 18 of one worker, 18.96 h into that fleet. What is published for it instead is its root constructor and seed, which rebuild the logged 139-gate depth-3 starting circuit exactly, and the worker’s untouched log of every chunk from there to 88 gates. That distinction changes nothing about correctness: every circuit any of it produced is machine-verified against MixColumns rebuilt from $\text{GF}(2^8)$, and the method is fully specified and reproducible from the accompanying code either way.

6 Scope and claims

- **Not optimality.** Minimum 2-input-XOR circuit size (the Shortest Linear Program problem) is NP-hard; no count here is proven minimal, and the certificates of Section 4 are local.
- **Comparisons.** All counts are for the model and convention of Section 1; the baselines [2, 3, 4, 5, 6, 7, 8, 9] were source-checked, and a source-by-source audit, with a dated Corrections section, is in the artifact repository’s `PRIOR_ART.md`. We claim the smallest counts at depths 3, 4, 5 and 6, and no improvement at unconstrained depth: the published floor is 88 and stays 88.
- **Priority. 88 is Jean’s count** [9], posted first; none of our four 88s is a count record. The one at depth 7 ties that point independently; the one at depth 6 improves the depth at that count; the ones at depths 5 and 8 are derived from Jean’s circuit, credited, and reported as derived wherever they appear. The depths quoted for [9] and [8] — 7 and 9 — are ours, not theirs, and are treated identically: see Table 1, note ‡.
- **Earlier results.** This note supersedes the project’s 98 at depth 3, 91 at depth 6 and 89 at depth 10, kept in the artifact repository.

Version history. **v3**, 29 July 2026: adds the 88 at depth 6 (the first 88 here found from scratch) and the derived 88 at depth 5, both frontier statements, the depth-obstruction datum, and the two new $k \leq 3/k = 2$ shells. Two *observations* of v2 are retracted, both refuted by these circuits: that no 88-gate mask set of build depth ≤ 6 had been seen, and that the best-certified 88-gate shells and this project’s Jean-independent 88 were disjoint sets (the depth-6

88 is now in both). No *claim* of v1 or v2 is withdrawn. **v2**, 27 July 2026: adds the 88 at depth 7 and its derived depth-8 companion, the exact neighbourhood certificates, the ρ^2 symmetry results, the v2 engine measurements and dated reproduction times; no claim of v1 is withdrawn. **v1**, July 2026 (the 97, 92 and 89 only), remains available in the repository history and in the Zenodo version archive under the concept DOI 10.5281/zenodo.21299092.

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A Circuit mixcolumns_97gates_depth3 (97 gates, depth 3)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e4c4e678c4b7c22d889c6e060b43859c111b82b953d6d9b43c49c079c0545b56

s32 = s0 $\oplus$ s8	s36 = s8 $\oplus$ s16	s40 = s11 $\oplus$ s20	s44 = s1 $\oplus$ s24
s33 = s5 $\oplus$ s13	s37 = s9 $\oplus$ s17	s41 = s12 $\oplus$ s20	s45 = s1 $\oplus$ s25
s34 = s6 $\oplus$ s14	s38 = s10 $\oplus$ s18	s42 = s6 $\oplus$ s22	s46 = s9 $\oplus$ s25
s35 = s7 $\oplus$ s15	s39 = s11 $\oplus$ s19	s43 = s15 $\oplus$ s23	s47 = s2 $\oplus$ s26

s48 = s11 $\oplus$ s26	s69 = s44 $\oplus$ s37	s90 = s57 $\oplus$ s0	s111 = s86 $\oplus$ s89
s49 = s2 $\oplus$ s27	s70 = s32 $\oplus$ s46	s91 = s57 $\oplus$ s3	s112 = s93 $\oplus$ s64
s50 = s3 $\oplus$ s27	s71 = s46 $\oplus$ s10	s92 = s57 $\oplus$ s15	s113 = s67 $\oplus$ s58
s51 = s4 $\oplus$ s27	s72 = s46 $\oplus$ s16	s93 = s34 $\oplus$ s58	s114 = s94 $\oplus$ s72
s52 = s4 $\oplus$ s28	s73 = s45 $\oplus$ s17	s94 = s44 $\oplus$ s58	s115 = s75 $\oplus$ s71
s53 = s12 $\oplus$ s28	s74 = s45 $\oplus$ s38	s95 = s48 $\oplus$ s58	s116 = s95 $\oplus$ s79
s54 = s13 $\oplus$ s29	s75 = s47 $\oplus$ s37	s96 = s53 $\oplus$ s58	s117 = s96 $\oplus$ s81
s55 = s21 $\oplus$ s29	s76 = s47 $\oplus$ s39	s97 = s68 $\oplus$ s35	s118 = s87 $\oplus$ s62
s56 = s22 $\oplus$ s30	s77 = s46 $\oplus$ s26	s98 = s70 $\oplus$ s60	s119 = s89 $\oplus$ s34
s57 = s7 $\oplus$ s31	s78 = s49 $\oplus$ s35	s99 = s74 $\oplus$ s77	s120 = s92 $\oplus$ s56
s58 = s23 $\oplus$ s31	s79 = s50 $\oplus$ s18	s100 = s78 $\oplus$ s61	s121 = s90 $\oplus$ s36
s59 = s33 $\oplus$ s14	s80 = s50 $\oplus$ s38	s101 = s63 $\oplus$ s83	s122 = s69 $\oplus$ s90
s60 = s35 $\oplus$ s17	s81 = s51 $\oplus$ s19	s102 = s87 $\oplus$ s85	s123 = s74 $\oplus$ s2
s61 = s39 $\oplus$ s10	s82 = s51 $\oplus$ s41	s103 = s59 $\oplus$ s56	s124 = s76 $\oplus$ s91
s62 = s41 $\oplus$ s5	s83 = s53 $\oplus$ s3	s104 = s93 $\oplus$ s15	s125 = s82 $\oplus$ s91
s63 = s35 $\oplus$ s40	s84 = s52 $\oplus$ s40	s105 = s65 $\oplus$ s67	s126 = s85 $\oplus$ s33
s64 = s42 $\oplus$ s7	s85 = s52 $\oplus$ s21	s106 = s65 $\oplus$ s73	s127 = s86 $\oplus$ s59
s65 = s36 $\oplus$ s43	s86 = s42 $\oplus$ s54	s107 = s75 $\oplus$ s18	s128 = s88 $\oplus$ s64
s66 = s43 $\oplus$ s19	s87 = s53 $\oplus$ s54	s108 = s80 $\oplus$ s66	
s67 = s32 $\oplus$ s24	s88 = s43 $\oplus$ s56	s109 = s84 $\oplus$ s66	
s68 = s36 $\oplus$ s24	s89 = s55 $\oplus$ s30	s110 = s62 $\oplus$ s55	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128]

B Circuit mixcolumns_92gates_depth4 (92 gates, depth 4)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e79bfccf5322fe8933242b47b53e94e73163615c70d5c45c9721e27ac86fd97d

s32 = s10 $\oplus$ s26	s56 = s15 $\oplus$ s31	s80 = s45 $\oplus$ s49	s104 = s63 $\oplus$ s49
s33 = s12 $\oplus$ s20	s57 = s43 $\oplus$ s44	s81 = s21 $\oplus$ s33	s105 = s64 $\oplus$ s50
s34 = s14 $\oplus$ s30	s58 = s34 $\oplus$ s53	s82 = s67 $\oplus$ s81	s106 = s64 $\oplus$ s69
s35 = s6 $\oplus$ s14	s59 = s5 $\oplus$ s54	s83 = s57 $\oplus$ s66	s107 = s23 $\oplus$ s75
s36 = s11 $\oplus$ s28	s60 = s0 $\oplus$ s37	s84 = s58 $\oplus$ s77	s108 = s71 $\oplus$ s80
s37 = s8 $\oplus$ s16	s61 = s38 $\oplus$ s52	s85 = s42 $\oplus$ s81	s109 = s46 $\oplus$ s72
s38 = s7 $\oplus$ s31	s62 = s6 $\oplus$ s40	s86 = s59 $\oplus$ s67	s110 = s102 $\oplus$ s66
s39 = s17 $\oplus$ s25	s63 = s32 $\oplus$ s56	s87 = s59 $\oplus$ s40	s111 = s18 $\oplus$ s101
s40 = s13 $\oplus$ s21	s64 = s36 $\oplus$ s56	s88 = s10 $\oplus$ s73	s112 = s89 $\oplus$ s78
s41 = s2 $\oplus$ s26	s65 = s37 $\oplus$ s56	s89 = s45 $\oplus$ s79	s113 = s63 $\oplus$ s108
s42 = s5 $\oplus$ s29	s66 = s1 $\oplus$ s48	s90 = s23 $\oplus$ s68	s114 = s108 $\oplus$ s55
s43 = s16 $\oplus$ s24	s67 = s50 $\oplus$ s53	s91 = s62 $\oplus$ s52	s115 = s93 $\oplus$ s65
s44 = s9 $\oplus$ s25	s68 = s35 $\oplus$ s56	s92 = s58 $\oplus$ s62	s116 = s47 $\oplus$ s109
s45 = s18 $\oplus$ s26	s69 = s3 $\oplus$ s38	s93 = s60 $\oplus$ s76	s117 = s52 $\oplus$ s107
s46 = s0 $\oplus$ s17	s70 = s51 $\oplus$ s54	s94 = s35 $\oplus$ s77	s118 = s96 $\oplus$ s104
s47 = s1 $\oplus$ s9	s71 = s19 $\oplus$ s48	s95 = s15 $\oplus$ s61	s119 = s33 $\oplus$ s106
s48 = s23 $\oplus$ s31	s72 = s24 $\oplus$ s38	s96 = s69 $\oplus$ s74	s120 = s71 $\oplus$ s105
s49 = s3 $\oplus$ s27	s73 = s39 $\oplus$ s41	s97 = s65 $\oplus$ s72	s121 = s103 $\oplus$ s51
s50 = s4 $\oplus$ s20	s74 = s41 $\oplus$ s55	s98 = s38 $\oplus$ s60	s122 = s97 $\oplus$ s100
s51 = s12 $\oplus$ s27	s75 = s34 $\oplus$ s38	s99 = s70 $\oplus$ s71	s123 = s90 $\oplus$ s75
s52 = s22 $\oplus$ s30	s76 = s43 $\oplus$ s48	s100 = s57 $\oplus$ s46	
s53 = s13 $\oplus$ s29	s77 = s22 $\oplus$ s42	s101 = s39 $\oplus$ s78	
s54 = s4 $\oplus$ s28	s78 = s41 $\oplus$ s44	s102 = s39 $\oplus$ s65	
s55 = s11 $\oplus$ s19	s79 = s10 $\oplus$ s47	s103 = s69 $\oplus$ s50	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 122, 89, 118, 119, 82, 84, 90, 115, 110, 111, 113, 120, 85, 91, 117, 93, 83, 88, 114, 99, 86, 92, 95, 98, 116, 112, 96, 121, 87, 94, 123]

C Circuit mixcolumns_89gates_depth5 (89 gates, depth 5)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

c3b972af282d25e8313a510f5cf840472401c6268bd28cda889593fc3379f064

s32 = s9 $\oplus$ s17	s55 = s21 $\oplus$ s39	s78 = s55 $\oplus$ s66	s101 = s60 $\oplus$ s50
s33 = s19 $\oplus$ s27	s56 = s5 $\oplus$ s46	s79 = s37 $\oplus$ s57	s102 = s39 $\oplus$ s101
s34 = s1 $\oplus$ s25	s57 = s14 $\oplus$ s42	s80 = s35 $\oplus$ s56	s103 = s34 $\oplus$ s87
s35 = s21 $\oplus$ s29	s58 = s15 $\oplus$ s37	s81 = s56 $\oplus$ s66	s104 = s59 $\oplus$ s89
s36 = s12 $\oplus$ s28	s59 = s11 $\oplus$ s33	s82 = s55 $\oplus$ s42	s105 = s76 $\oplus$ s70
s37 = s22 $\oplus$ s30	s60 = s11 $\oplus$ s43	s83 = s70 $\oplus$ s72	s106 = s92 $\oplus$ s65
s38 = s3 $\oplus$ s27	s61 = s0 $\oplus$ s52	s84 = s43 $\oplus$ s71	s107 = s96 $\oplus$ s98
s39 = s4 $\oplus$ s28	s62 = s33 $\oplus$ s45	s85 = s54 $\oplus$ s67	s108 = s36 $\oplus$ s93
s40 = s9 $\oplus$ s25	s63 = s38 $\oplus$ s52	s86 = s54 $\oplus$ s49	s109 = s95 $\oplus$ s48
s41 = s13 $\oplus$ s29	s64 = s43 $\oplus$ s51	s87 = s17 $\oplus$ s64	s110 = s1 $\oplus$ s92
s42 = s5 $\oplus$ s13	s65 = s8 $\oplus$ s45	s88 = s57 $\oplus$ s67	s111 = s99 $\oplus$ s46
s43 = s15 $\oplus$ s23	s66 = s36 $\oplus$ s41	s89 = s60 $\oplus$ s47	s112 = s98 $\oplus$ s75
s44 = s6 $\oplus$ s22	s67 = s41 $\oplus$ s44	s90 = s62 $\oplus$ s53	s113 = s89 $\oplus$ s90
s45 = s7 $\oplus$ s23	s68 = s34 $\oplus$ s53	s91 = s71 $\oplus$ s74	s114 = s108 $\oplus$ s99
s46 = s12 $\oplus$ s20	s69 = s24 $\oplus$ s45	s92 = s24 $\oplus$ s61	s115 = s110 $\oplus$ s112
s47 = s10 $\oplus$ s18	s70 = s40 $\oplus$ s47	s93 = s38 $\oplus$ s62	s116 = s3 $\oplus$ s104
s48 = s2 $\oplus$ s26	s71 = s31 $\oplus$ s49	s94 = s58 $\oplus$ s52	s117 = s90 $\oplus$ s109
s49 = s6 $\oplus$ s14	s72 = s26 $\oplus$ s34	s95 = s59 $\oplus$ s63	s118 = s108 $\oplus$ s101
s50 = s19 $\oplus$ s20	s73 = s32 $\oplus$ s48	s96 = s64 $\oplus$ s69	s119 = s87 $\oplus$ s112
s51 = s8 $\oplus$ s16	s74 = s44 $\oplus$ s45	s97 = s61 $\oplus$ s51	s120 = s32 $\oplus$ s110
s52 = s7 $\oplus$ s31	s75 = s40 $\oplus$ s51	s98 = s0 $\oplus$ s65	
s53 = s2 $\oplus$ s18	s76 = s18 $\oplus$ s73	s99 = s4 $\oplus$ s63	
s54 = s30 $\oplus$ s35	s77 = s10 $\oplus$ s68	s100 = s58 $\oplus$ s74	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[96, 119, 83, 113, 118, 78, 79, 84, 107, 103, 76, 116, 102, 80, 85, 91, 106, 115, 105, 117, 114, 81, 86, 94, 97, 120, 77, 109, 111, 82, 88, 100]

D Circuit mixcolumns_88gates_depth5 (88 gates, depth 5; derived, shallowest 88 here)

Status: derived from published work. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), so this is not an independent construction and Jean is credited for the material it descends from. It is the shallowest 88 here and six gates below the published depth-5 94 [3]; it shares 75 of its 88 masks with Appendix C (Jaccard 0.735), the depth-5 89's own basin reached at 88 gates. 88 is Jean's count, not a new one.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

91d45c57d638dc5832a23eaf7721ba80d07bc0566769c47b56bd5566ae4eabff

s32 = s13 $\oplus$ s29	s47 = s6 $\oplus$ s22	s62 = s40 $\oplus$ s51	s77 = s51 $\oplus$ s0
s33 = s21 $\oplus$ s29	s48 = s6 $\oplus$ s14	s63 = s41 $\oplus$ s21	s78 = s51 $\oplus$ s56
s34 = s22 $\oplus$ s30	s49 = s11 $\oplus$ s19	s64 = s42 $\oplus$ s45	s79 = s51 $\oplus$ s60
s35 = s15 $\oplus$ s23	s50 = s7 $\oplus$ s23	s65 = s42 $\oplus$ s52	s80 = s52 $\oplus$ s77
s36 = s9 $\oplus$ s17	s51 = s7 $\oplus$ s31	s66 = s43 $\oplus$ s26	s81 = s53 $\oplus$ s63
s37 = s12 $\oplus$ s20	s52 = s8 $\oplus$ s16	s67 = s43 $\oplus$ s61	s82 = s53 $\oplus$ s59
s38 = s12 $\oplus$ s28	s53 = s32 $\oplus$ s38	s68 = s44 $\oplus$ s14	s83 = s54 $\oplus$ s55
s39 = s3 $\oplus$ s27	s54 = s32 $\oplus$ s47	s69 = s44 $\oplus$ s63	s84 = s54 $\oplus$ s68
s40 = s3 $\oplus$ s19	s55 = s33 $\oplus$ s30	s70 = s45 $\oplus$ s11	s85 = s56 $\oplus$ s72
s41 = s4 $\oplus$ s28	s56 = s34 $\oplus$ s15	s71 = s46 $\oplus$ s26	s86 = s57 $\oplus$ s20
s42 = s9 $\oplus$ s25	s57 = s35 $\oplus$ s49	s72 = s47 $\oplus$ s50	s87 = s58 $\oplus$ s17
s43 = s2 $\oplus$ s18	s58 = s35 $\oplus$ s52	s73 = s48 $\oplus$ s55	s88 = s59 $\oplus$ s33
s44 = s5 $\oplus$ s13	s59 = s37 $\oplus$ s5	s74 = s48 $\oplus$ s31	s89 = s61 $\oplus$ s38
s45 = s10 $\oplus$ s18	s60 = s39 $\oplus$ s4	s75 = s50 $\oplus$ s8	s90 = s64 $\oplus$ s71
s46 = s1 $\oplus$ s25	s61 = s40 $\oplus$ s50	s76 = s50 $\oplus$ s58	s91 = s66 $\oplus$ s36

s92 = s66 $\oplus$ s10	s100 = s77 $\oplus$ s24	s108 = s94 $\oplus$ s57	s116 = s109 $\oplus$ s35
s93 = s68 $\oplus$ s34	s101 = s79 $\oplus$ s89	s109 = s94 $\oplus$ s67	s117 = s111 $\oplus$ s112
s94 = s70 $\oplus$ s39	s102 = s79 $\oplus$ s37	s110 = s98 $\oplus$ s99	s118 = s111 $\oplus$ s87
s95 = s71 $\oplus$ s92	s103 = s86 $\oplus$ s41	s111 = s98 $\oplus$ s65	s119 = s112 $\oplus$ s36
s96 = s72 $\oplus$ s74	s104 = s86 $\oplus$ s89	s112 = s100 $\oplus$ s1	
s97 = s74 $\oplus$ s35	s105 = s87 $\oplus$ s46	s113 = s100 $\oplus$ s75	
s98 = s75 $\oplus$ s0	s106 = s91 $\oplus$ s64	s114 = s107 $\oplus$ s109	
s99 = s76 $\oplus$ s24	s107 = s92 $\oplus$ s62	s115 = s107 $\oplus$ s70	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[99, 118, 90, 116, 104, 81, 93, 97, 110, 105, 91, 108, 103, 88, 83, 96, 113, 117, 106, 114, 101, 82, 73, 78, 80, 119, 95, 115, 102, 69, 84, 85]

E Circuit mixcolumns_88gates_depth6 (88 gates, depth 6; improves the frontier at depth 6)

Status: found from scratch on this project's own lineage, with no imported circuit at any step. 88 is not a new gate count — it is Jean's [9], who has priority. What this circuit improves is the depth at that count: one level shallower than Jean's 88, and four gates below the published depth-6 92 [4]. Its remove- ≤ 3 shell is exhaustively empty (Section 4).

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

462a149979fe0ab39173e6aa70c1728fe55cb24b380ea0b2241ffaa6301a8c4c

s32 = s14 $\oplus$ s22	s55 = s40 $\oplus$ s12	s78 = s57 $\oplus$ s69	s101 = s80 $\oplus$ s86
s33 = s13 $\oplus$ s29	s56 = s41 $\oplus$ s51	s79 = s58 $\oplus$ s2	s102 = s81 $\oplus$ s8
s34 = s21 $\oplus$ s29	s57 = s41 $\oplus$ s6	s80 = s58 $\oplus$ s12	s103 = s84 $\oplus$ s38
s35 = s14 $\oplus$ s30	s58 = s42 $\oplus$ s43	s81 = s59 $\oplus$ s37	s104 = s84 $\oplus$ s90
s36 = s20 $\oplus$ s28	s59 = s42 $\oplus$ s46	s82 = s59 $\oplus$ s61	s105 = s85 $\oplus$ s91
s37 = s0 $\oplus$ s24	s60 = s43 $\oplus$ s3	s83 = s60 $\oplus$ s36	s106 = s85 $\oplus$ s27
s38 = s17 $\oplus$ s25	s61 = s44 $\oplus$ s24	s84 = s61 $\oplus$ s16	s107 = s87 $\oplus$ s55
s39 = s2 $\oplus$ s26	s62 = s44 $\oplus$ s10	s85 = s62 $\oplus$ s79	s108 = s88 $\oplus$ s92
s40 = s4 $\oplus$ s20	s63 = s44 $\oplus$ s11	s86 = s63 $\oplus$ s83	s109 = s92 $\oplus$ s49
s41 = s5 $\oplus$ s13	s64 = s45 $\oplus$ s27	s87 = s64 $\oplus$ s3	s110 = s94 $\oplus$ s96
s42 = s15 $\oplus$ s23	s65 = s45 $\oplus$ s49	s88 = s65 $\oplus$ s35	s111 = s94 $\oplus$ s54
s43 = s11 $\oplus$ s19	s66 = s45 $\oplus$ s46	s89 = s65 $\oplus$ s42	s112 = s96 $\oplus$ s52
s44 = s7 $\oplus$ s23	s67 = s47 $\oplus$ s63	s90 = s66 $\oplus$ s0	s113 = s98 $\oplus$ s51
s45 = s7 $\oplus$ s31	s68 = s48 $\oplus$ s57	s91 = s67 $\oplus$ s87	s114 = s103 $\oplus$ s111
s46 = s8 $\oplus$ s16	s69 = s50 $\oplus$ s22	s92 = s69 $\oplus$ s44	s115 = s103 $\oplus$ s9
s47 = s18 $\oplus$ s26	s70 = s50 $\oplus$ s34	s93 = s70 $\oplus$ s48	s116 = s105 $\oplus$ s54
s48 = s32 $\oplus$ s33	s71 = s52 $\oplus$ s59	s94 = s72 $\oplus$ s1	s117 = s107 $\oplus$ s86
s49 = s32 $\oplus$ s15	s72 = s53 $\oplus$ s9	s95 = s72 $\oplus$ s18	s118 = s111 $\oplus$ s37
s50 = s35 $\oplus$ s6	s73 = s53 $\oplus$ s10	s96 = s73 $\oplus$ s47	s119 = s115 $\oplus$ s81
s51 = s36 $\oplus$ s21	s74 = s54 $\oplus$ s60	s97 = s73 $\oplus$ s25	
s52 = s38 $\oplus$ s1	s75 = s55 $\oplus$ s28	s98 = s75 $\oplus$ s33	
s53 = s39 $\oplus$ s17	s76 = s56 $\oplus$ s34	s99 = s75 $\oplus$ s80	
s54 = s39 $\oplus$ s45	s77 = s56 $\oplus$ s40	s100 = s77 $\oplus$ s98	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[82, 119, 110, 106, 101, 113, 78, 108, 102, 71, 95, 116, 99, 100, 93, 89, 104, 114, 97, 91, 117, 76, 70, 88, 90, 118, 112, 74, 107, 77, 68, 109]

F Circuit mixcolumns_88gates_depth7 (88 gates, depth 7; ties the published floor)

Status: the same (depth, count) point as Jean [9], reached independently on this project's own lineage; 61 of 88 internal masks are shared. It ties that point and does not beat it, and Jean has priority. It is dominated by Appendix E, and kept because independent replication of a

published point is worth archiving.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

76bac50ab0e4a7dd54578e77f57f6d08e96a8803aff72dba2c3f5fadaaa06ee1

s32 = s13 $\oplus$ s29	s55 = s34 $\oplus$ s15	s78 = s52 $\oplus$ s64	s101 = s87 $\oplus$ s37
s33 = s21 $\oplus$ s29	s56 = s35 $\oplus$ s20	s79 = s53 $\oplus$ s63	s102 = s90 $\oplus$ s61
s34 = s22 $\oplus$ s30	s57 = s35 $\oplus$ s8	s80 = s53 $\oplus$ s59	s103 = s90 $\oplus$ s40
s35 = s15 $\oplus$ s23	s58 = s36 $\oplus$ s2	s81 = s55 $\oplus$ s68	s104 = s94 $\oplus$ s38
s36 = s17 $\oplus$ s25	s59 = s37 $\oplus$ s5	s82 = s57 $\oplus$ s16	s105 = s94 $\oplus$ s40
s37 = s12 $\oplus$ s20	s60 = s39 $\oplus$ s1	s83 = s58 $\oplus$ s61	s106 = s95 $\oplus$ s60
s38 = s12 $\oplus$ s28	s61 = s39 $\oplus$ s51	s84 = s59 $\oplus$ s33	s107 = s96 $\oplus$ s44
s39 = s9 $\oplus$ s25	s62 = s41 $\oplus$ s50	s85 = s60 $\oplus$ s17	s108 = s96 $\oplus$ s36
s40 = s2 $\oplus$ s18	s63 = s42 $\oplus$ s21	s86 = s62 $\oplus$ s76	s109 = s97 $\oplus$ s108
s41 = s3 $\oplus$ s27	s64 = s43 $\oplus$ s14	s87 = s62 $\oplus$ s4	s110 = s98 $\oplus$ s49
s42 = s4 $\oplus$ s28	s65 = s43 $\oplus$ s63	s88 = s64 $\oplus$ s34	s111 = s100 $\oplus$ s105
s43 = s5 $\oplus$ s13	s66 = s44 $\oplus$ s45	s89 = s66 $\oplus$ s86	s112 = s104 $\oplus$ s87
s44 = s15 $\oplus$ s31	s67 = s45 $\oplus$ s58	s90 = s67 $\oplus$ s85	s113 = s104 $\oplus$ s71
s45 = s10 $\oplus$ s26	s68 = s46 $\oplus$ s49	s91 = s68 $\oplus$ s70	s114 = s106 $\oplus$ s108
s46 = s6 $\oplus$ s22	s69 = s47 $\oplus$ s54	s92 = s70 $\oplus$ s35	s115 = s107 $\oplus$ s74
s47 = s6 $\oplus$ s14	s70 = s47 $\oplus$ s31	s93 = s71 $\oplus$ s42	s116 = s107 $\oplus$ s110
s48 = s11 $\oplus$ s19	s71 = s48 $\oplus$ s56	s94 = s72 $\oplus$ s3	s117 = s107 $\oplus$ s57
s49 = s7 $\oplus$ s23	s72 = s49 $\oplus$ s19	s95 = s73 $\oplus$ s74	s118 = s111 $\oplus$ s41
s50 = s7 $\oplus$ s31	s73 = s49 $\oplus$ s16	s96 = s73 $\oplus$ s0	s119 = s111 $\oplus$ s66
s51 = s18 $\oplus$ s26	s74 = s50 $\oplus$ s24	s97 = s82 $\oplus$ s9	
s52 = s32 $\oplus$ s46	s75 = s50 $\oplus$ s55	s98 = s82 $\oplus$ s24	
s53 = s32 $\oplus$ s38	s76 = s51 $\oplus$ s72	s99 = s85 $\oplus$ s97	
s54 = s33 $\oplus$ s30	s77 = s52 $\oplus$ s54	s100 = s86 $\oplus$ s48	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[110, 109, 103, 119, 113, 79, 88, 92, 115, 99, 83, 89, 93, 84, 77, 91, 116, 106, 67, 100, 112, 80, 69, 75, 117, 114, 102, 118, 101, 65, 78, 81]

G Circuit mixcolumns_88gates_depth8 (88 gates, depth 8; derived, non-frontier)

Status: derived from published work, and not a frontier point. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), and it is dominated by every other 88 here (same count, greater depth). It is listed as a verified distinct construction, not as a claim.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

3a23e2f2b2f3835cab74ba23800088bb03ab4ee3f076f2b9a826c78cd84f8ef1

s32 = s13 $\oplus$ s29	s51 = s32 $\oplus$ s37	s70 = s51 $\oplus$ s58	s89 = s76 $\oplus$ s44
s33 = s21 $\oplus$ s29	s52 = s32 $\oplus$ s47	s71 = s52 $\oplus$ s53	s90 = s77 $\oplus$ s41
s34 = s22 $\oplus$ s30	s53 = s33 $\oplus$ s30	s72 = s52 $\oplus$ s60	s91 = s81 $\oplus$ s33
s35 = s15 $\oplus$ s23	s54 = s34 $\oplus$ s15	s73 = s54 $\oplus$ s64	s92 = s81 $\oplus$ s51
s36 = s9 $\oplus$ s17	s55 = s35 $\oplus$ s8	s74 = s55 $\oplus$ s16	s93 = s85 $\oplus$ s89
s37 = s12 $\oplus$ s28	s56 = s38 $\oplus$ s26	s75 = s56 $\oplus$ s36	s94 = s85 $\oplus$ s40
s38 = s2 $\oplus$ s18	s57 = s39 $\oplus$ s50	s76 = s56 $\oplus$ s10	s95 = s86 $\oplus$ s35
s39 = s3 $\oplus$ s27	s58 = s40 $\oplus$ s21	s77 = s57 $\oplus$ s12	s96 = s87 $\oplus$ s49
s40 = s4 $\oplus$ s28	s59 = s42 $\oplus$ s45	s78 = s59 $\oplus$ s75	s97 = s87 $\oplus$ s1
s41 = s4 $\oplus$ s20	s60 = s43 $\oplus$ s14	s79 = s59 $\oplus$ s63	s98 = s88 $\oplus$ s46
s42 = s9 $\oplus$ s25	s61 = s43 $\oplus$ s58	s80 = s60 $\oplus$ s34	s99 = s94 $\oplus$ s77
s43 = s5 $\oplus$ s13	s62 = s45 $\oplus$ s19	s81 = s61 $\oplus$ s41	s100 = s95 $\oplus$ s62
s44 = s15 $\oplus$ s31	s63 = s46 $\oplus$ s26	s82 = s63 $\oplus$ s76	s101 = s95 $\oplus$ s20
s45 = s10 $\oplus$ s18	s64 = s47 $\oplus$ s49	s83 = s64 $\oplus$ s66	s102 = s96 $\oplus$ s16
s46 = s1 $\oplus$ s25	s65 = s48 $\oplus$ s53	s84 = s66 $\oplus$ s35	s103 = s96 $\oplus$ s8
s47 = s6 $\oplus$ s22	s66 = s48 $\oplus$ s31	s85 = s67 $\oplus$ s19	s104 = s97 $\oplus$ s36
s48 = s6 $\oplus$ s14	s67 = s49 $\oplus$ s3	s86 = s67 $\oplus$ s11	s105 = s100 $\oplus$ s89
s49 = s7 $\oplus$ s23	s68 = s50 $\oplus$ s24	s87 = s68 $\oplus$ s0	s106 = s100 $\oplus$ s39
s50 = s7 $\oplus$ s31	s69 = s50 $\oplus$ s54	s88 = s74 $\oplus$ s17	s107 = s101 $\oplus$ s94

$s_{108} = s_{101} \oplus s_{37}$	$s_{112} = s_{106} \oplus s_{38}$	$s_{116} = s_{113} \oplus s_{103}$
$s_{109} = s_{102} \oplus s_{44}$	$s_{113} = s_{109} \oplus s_{68}$	$s_{117} = s_{113} \oplus s_{55}$
$s_{110} = s_{102} \oplus s_{68}$	$s_{114} = s_{110} \oplus s_{42}$	$s_{118} = s_{114} \oplus s_{97}$
$s_{111} = s_{106} \oplus s_{86}$	$s_{115} = s_{112} \oplus s_{93}$	$s_{119} = s_{114} \oplus s_{88}$

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[116, 119, 79, 112, 108, 70, 80, 84, 109, 98, 75, 111, 107, 92, 71, 83, 103, 118, 78, 115, 99, 91, 65,
69, 117, 104, 82, 105, 90, 61, 72, 73]